FRIDAY 13 NOVEMBER 2026 · 09:00–10:00 · GROUP 07

Linear Algebra — Week 8 tutorial

The fields F_2 and F_3, vector spaces over them, and isomorphisms

1 · Example
Watch one worked example and copy it down.

2 · Your turn
Do one like it with your neighbour.

3 · Compare
We check it together; write one takeaway.

Key ideas

1. **Small fields.** $\mathbb{F}_2 = \{0, 1\}$ with arithmetic mod 2 (so $1 + 1 = 0$); $\mathbb{F}_3 = \{0, 1, 2\}$ with arithmetic mod 3. Every non-zero element has a multiplicative inverse.
2. **Vectors over $\mathbb{F}_p$.** $(\mathbb{F}_p)^n$ has entries *and scalars* from $\mathbb{F}_p$, so it contains exactly p^n vectors.
3. **Polynomial spaces.** $P_n(K)$ = polynomials of degree at most n with coefficients in K; basis $1, x, \ldots, x^n$; dimension $n + 1$.
4. **Isomorphism.** A linear map that is bijective. If $\dim_K V = n$ then $V \cong K^n$; over the same field, same (finite) dimension $\iff$ isomorphic.

Recall

No notes, 2 minutes on your own, then compare with your neighbour.

1. Write down the two conditions for a map $T : V \to W$ to be linear.

2. $T : \mathbb{R}^3 \to \mathbb{R}^2$ is linear and surjective. What is its nullity?

3. True or false: if T is linear and $\ker T = \{0\}$, then T is injective.

After the tutorial

- **From today:** Problem Sheet 7 Q2 (subspaces of $(\mathbb{F}_2)^n$ — follows on from Round 1), Q5(c) (d) and Q7 (isomorphisms).
- **Then:** Q3, Q4, and the Numbas quiz.
- **Hand-in:** Q1, by noon on Wednesday 18 November.

Solutions to the additional questions are on Blackboard with Problem Sheet 7.

Adapted from Linear Algebra Problem Sheet 7 (University of Bristol, 2026/27) for tutorial use. No solutions are included.

Round 1 · Arithmetic in $\mathbb{F}_2$ and $\mathbb{F}_3$

EXAMPLE — we do this one together

(i) Write the addition and multiplication tables of $\mathbb{F}_2$.

(ii) List the elements of $(\mathbb{F}_2)^2$. What is $v + v$ for a vector v in this space?

YOUR TURN — with your neighbour

(a) Fill in the addition and multiplication tables of $\mathbb{F}_3 = \{0, 1, 2\}$.

(b) In $\mathbb{F}_3$: what is -1? What is 2^{-1}?

(c) Solve $2x + 1 = 0$ in $\mathbb{F}_3$.

(d) How many vectors are there in $(\mathbb{F}_3)^2$?

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Finished early? Is $\{0, 1, 2, 3\}$ with arithmetic mod 4 a field? (Look at 2×2.)

Takeaway — in your own words:

Challenge — harder; work on the back of this page (Linear PS7 Q2(a)(b))

Let V be a vector space over $\mathbb{F}_2$.

(a) Which part of the subspace test never needs checking in V? Why?

(b) Hence find the number of subspaces of $(\mathbb{F}_2)^2$, and then of $(\mathbb{F}_2)^3$.

Round 2 · Polynomials over $\mathbb{F}_2$: counting, bases, and a surprise

EXAMPLE — we do this one together

$P_1(\mathbb{F}_2)$ is the set of polynomials $a_0 + a_1 x$ with $a_0, a_1 \in \mathbb{F}_2$.

(i) List its elements. (ii) Give a basis. (iii) Is $\{1,\ 1+x\}$ also a basis?

YOUR TURN — with your neighbour

Now take $P_2(\mathbb{F}_2)$: polynomials $a_0 + a_1 x + a_2 x^2$ with coefficients in $\mathbb{F}_2$.

(a) How many elements does it have? List them.

(b) Write down two *different* bases.

(c) Is $\{\,1+x,\ \ x+x^2,\ \ 1+x^2\,\}$ linearly independent over $\mathbb{F}_2$? (Try adding all three.)

Finished early? How many elements does $P_3(\mathbb{F}_2)$ have? Which elements of $P_2(\mathbb{F}_2)$ are not in $\mathrm{Span}(1+x,\ x+x^2)$?

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*Finished early? Write down an explicit isomorphism between the two isomorphic spaces. **Challenge:** prove that $\mathbb{R}^2$ and $\mathbb{R}^3$ are not isomorphic.*

Takeaway — in your own words:

Exit ticket · on your own, 3 min

Least sure about? Circle: Round 1 | Round 2 | Round 3

1. In $\mathbb{F}_3$, compute $2 + 2$ and 2×2.
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(b) Write down two *different* bases.

(c) Is $\{\,1+x,\ \ x+x^2,\ \ 1+x^2\,\}$ linearly independent over $\mathbb{F}_2$? (Try adding all three.)

Finished early? How many elements does $P_3(\mathbb{F}_2)$ have? Which elements of $P_2(\mathbb{F}_2)$ are not in $\mathrm{Span}(1+x,\ x+x^2)$?

Takeaway — in your own words:

Challenge — harder; work on the back of this page

(a) How many different bases does $P_1(\mathbb{F}_2)$ have?

(b) How many *ordered* bases (v_1, v_2, v_3) does $P_2(\mathbb{F}_2)$ have? (Choose $v_1 \neq 0$; then $v_2 \notin \mathrm{Span}(v_1)$; then $v_3 \notin \mathrm{Span}(v_1, v_2)$.)

Round 3 · Isomorphism: when are two spaces “the same”? if time

EXAMPLE — we do this one together (Linear PS7 Q5(a)(b))

Let $\varphi : \mathbb{R}^3 \to P_2$ send e_1 to 1, e_2 to x, and e_3 to x^2.

(a) If φ is linear, what is the output of a general $(a_1, a_2, a_3) \in \mathbb{R}^3$?

(b) Is φ an isomorphism? What do you need to prove?

YOUR TURN — with your neighbour

Fill in a basis and the dimension of each space *over the field shown*. Then decide: which of these spaces are isomorphic to each other?

Space	over	a basis	dim
$\mathbb{R}^2$	$\mathbb{R}$		
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*Finished early? Write down an explicit isomorphism between the two isomorphic spaces. **Challenge:** prove that $\mathbb{R}^2$ and $\mathbb{R}^3$ are not isomorphic.*

Takeaway — in your own words:

Exit ticket · on your own, 3 min

Least sure about? Circle: Round 1 | Round 2 | Round 3

1. In $\mathbb{F}_3$, compute $2 + 2$ and 2×2.
2. How many vectors are there in $(\mathbb{F}_2)^3$?
3. Is $P_3(\mathbb{R})$ isomorphic to $\mathbb{R}^4$? Give a one-line reason.

FRIDAY 13 NOVEMBER 2026 · 09:00–10:00 · GROUP 07

Linear Algebra — Week 8 tutorial

The fields F_2 and F_3, vector spaces over them, and isomorphisms

1 · Example
Watch one worked example and copy it down.

2 · Your turn
Do one like it with your neighbour.

3 · Compare
We check it together; write one takeaway.

Key ideas

1. **Small fields.** $\mathbb{F}_2 = \{0, 1\}$ with arithmetic mod 2 (so $1 + 1 = 0$); $\mathbb{F}_3 = \{0, 1, 2\}$ with arithmetic mod 3. Every non-zero element has a multiplicative inverse.
2. **Vectors over $\mathbb{F}_p$.** $(\mathbb{F}_p)^n$ has entries *and scalars* from $\mathbb{F}_p$, so it contains exactly p^n vectors.
3. **Polynomial spaces.** $P_n(K)$ = polynomials of degree at most n with coefficients in K; basis $1, x, \ldots, x^n$; dimension $n + 1$.
4. **Isomorphism.** A linear map that is bijective. If $\dim_K V = n$ then $V \cong K^n$; over the same field, same (finite) dimension $\iff$ isomorphic.

Recall

No notes, 2 minutes on your own, then compare with your neighbour.

1. Write down the two conditions for a map $T : V \to W$ to be linear.

2. $T : \mathbb{R}^3 \to \mathbb{R}^2$ is linear and surjective. What is its nullity?

3. True or false: if T is linear and $\ker T = \{0\}$, then T is injective.

After the tutorial

- **From today:** Problem Sheet 7 Q2 (subspaces of $(\mathbb{F}_2)^n$ — follows on from Round 1), Q5(c) (d) and Q7 (isomorphisms).
- **Then:** Q3, Q4, and the Numbas quiz.
- **Hand-in:** Q1, by noon on Wednesday 18 November.

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Round 1 · Arithmetic in $\mathbb{F}_2$ and $\mathbb{F}_3$

EXAMPLE — we do this one together

(i) Write the addition and multiplication tables of $\mathbb{F}_2$.

(ii) List the elements of $(\mathbb{F}_2)^2$. What is $v + v$ for a vector v in this space?

YOUR TURN — with your neighbour

(a) Fill in the addition and multiplication tables of $\mathbb{F}_3 = \{0, 1, 2\}$.

(b) In $\mathbb{F}_3$: what is -1? What is 2^{-1}?

(c) Solve $2x + 1 = 0$ in $\mathbb{F}_3$.

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Finished early? Is $\{0, 1, 2, 3\}$ with arithmetic mod 4 a field? (Look at 2×2.)

Takeaway — in your own words:

Challenge — harder; work on the back of this page (Linear PS7 Q2(a)(b))

Let V be a vector space over $\mathbb{F}_2$.

(a) Which part of the subspace test never needs checking in V? Why?

(b) Hence find the number of subspaces of $(\mathbb{F}_2)^2$, and then of $(\mathbb{F}_2)^3$.

Round 2 · Polynomials over $\mathbb{F}_2$: counting, bases, and a surprise

EXAMPLE — we do this one together

$P_1(\mathbb{F}_2)$ is the set of polynomials $a_0 + a_1 x$ with $a_0, a_1 \in \mathbb{F}_2$.

(i) List its elements. (ii) Give a basis. (iii) Is $\{1,\ 1+x\}$ also a basis?

YOUR TURN — with your neighbour

Now take $P_2(\mathbb{F}_2)$: polynomials $a_0 + a_1 x + a_2 x^2$ with coefficients in $\mathbb{F}_2$.

(a) How many elements does it have? List them.

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Exit ticket · on your own, 3 min

Least sure about? Circle: Round 1 | Round 2 | Round 3

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FRIDAY 13 NOVEMBER 2026 · 09:00–10:00 · GROUP 07

Linear Algebra — Week 8 tutorial

The fields F_2 and F_3, vector spaces over them, and isomorphisms

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No notes, 2 minutes on your own, then compare with your neighbour.

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